Moe Mirzaei (Mohammadreza Mirzaei)

SUMMARY

An engineer with a passion for innovation. I bring a strong commitment to hard work, creative problem-solving, and sharp analytical skills. I aim to merge my technical knowledge with my passion to make a meaningful impact.

CONTACT

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EDUCATION

Doctor of Philosophy, Engineering and Computing Systems 2026-Present
Pennsylvania State University, Harrisburg

Master's degree, Civil/Transportation Engineering (GPA: 4/4) 2023-2025
University of Florida, Gainesville

Bachelor's degree, Civil Engineering (GPA: 3.4/4) 2017-2021
Sharif University of Technology, Tehran

INTERESTS

- Intelligent Transportation Systems (ITS)
- Traffic Operations and Safety
- Artificial Intelligence and Data Science applications
- Computer Vision and Image Processing

SKILLS

Programming languages:
Proficient in **Python, R** Familiar with Julia, MATLAB

Code editors:
Visual Studio Code, RStudio, Jupyter Notebook, PyCharm

Database: MongoDB, SQL

OS: Windows, Linux (Ubuntu)

Software:
Microsoft Office (Proficient in Word, Excel, and PowerPoint)
Docker, SPSS, Mathcad, GAMS

Simulators: HCS, TSIS CORSIM, TRANSYT-7F, SwashSim, VISSIM, VISUM, ArcGIS Pro

Soft skills: Critical thinking, Problem-solving, Teamwork, Project management

LEADERSHIP

Director of Communication of Penn State ITE Oct 2026 – Present
Institute of Transportation Engineers student chapter, Pennsylvania State University

Vice President of Gator ITE Apr 2023 – Apr 2024
Institute of Transportation Engineers student chapter, University of Florida

President of Civil Engineering Club (Scientific Association) Jun 2019 - Aug 2020
Sharif University of Technology

LANGUAGES

English, Persian, and Azerbaijani

ACADEMIC EXPERIENCES

Graduate Research Assistant Aug 2026 – Present

Dr. Agnimitra Sengupta, Pennsylvania State University Harrisburg

- Research on traffic safety and operation, and AI applications in transportation engineering

Graduate Research Assistant (I-STREET Lab) Aug 2023 – May 2025

Dr. Lily Elefteriadou & Dr. Pruthvi Manjunatha, University of Florida Transportation Institute

Title: ***Evaluation of Gainesville Pedestrian-Bicyclists Connected Vehicle Pilot***

- Evaluated the passive pedestrian detection system, CV communication reliability, and user perceptions of the smartphone application by conducting focus groups and participant experiments.
- Worked with Roadside Units, On-board units, TravelSafely/Glance application, and Bosch cameras.
- Collaborated with several vendors (Yunex, Applied Information, Bosch, etc.) and FDOT.

Research (Remote) May 2022 – Apr 2023

Dr. Kara Kockelman, University of Texas at Austin

Title: *Inferring vehicle speeds, makes and models from video camera footage: Ready opportunities for transport data analysis.*

Research Assistant Aug 2020 – Sep 2021

Dr. Amir Samimi, Sharif University of Technology

➢ Researched Urban Freight Transportation types and algorithms for optimizing **Vehicle Routing Problems (VRP) and Pickup and Delivery Problems (PDP)**. Designed and developed a dashboard for solving several types of VRP with constraints using two algorithms (*Genetic Algorithm* and *Simulated Annealing*) in ***Python* and *R*** programming languages.

Teaching Assistant

- Transportation Engineering Dr. Pruthvi Manjunatha, University of Florida — Oct 2024
- Python Programming Dr. Elham Seifossadat, Sharif University of Technology — Feb 2020 – Oct 2021
- Engineering Probability and Statistics Dr. Hatef Abdoos, Sharif University of Technology — Sep 2019 – Dec 2019

WORK EXPERIENCES

ITS Traffic Engineer Jun 2025 – Mar 2026

AtkinsRealis, Florida Department of Transportation District Two (FDOT D2)

In this role, I supported the FDOT D2 on a range of ITS initiatives, including Connected Vehicle deployments, innovative Truck Parking Availability System (iTPAS), Rail Crossing Detection, Wrong Way Vehicle Detection, Video Analytics, and other emerging technology evaluations.

Transportation Data Analyst (Internship) Jul 2020 - Aug 2020

Roads and Transportation Information Management Center

1) Researched the information provided by traffic cameras in different countries.
2) Troubleshot the location and algorithm of travel time cameras.
3) Detected and troubleshooted traffic counters in three-ways.

CERTIFICATES

IMSA Traffic Signal Technician I Mar 2026 - Mar 2029

IMSA-International Municipal Signal Association *Credential ID AA_150495*

Advanced Temporary Traffic Control (TTC) Maintenance of Traffic (MOT) Oct 2025- Oct 2029

Florida Department of Transportation *Credential ID 649038*

NEMA Troubleshooting Program Oct 2025

East Coast Training Alliance